

**TrueLoad**

Real Data. Real Fast. Capacity Confirmed.

NEC 220.87

MEASURED DEMAND ANALYSIS

Service adequacy, backed by real data.

TrueLoad turns a year of the home's own utility meter data into a code-compliant **NEC 220.87 load calculation** — and a report ready to hand to the inspector. Prove the existing service can carry a heat pump, EV charger, or other new load **before anyone pays for a service upgrade**.

Licensed Electricians

Heat Pump Installers

EV Charger Installers

Residential Contractors

THE PROBLEM

Inspectors want documentation.

AHJs increasingly require a formal calculation before approving new loads on an existing service. A verbal assurance isn't enough.

Unneeded upgrades kill jobs.

A service upgrade adds \$5,000+ and weeks of delay. Recommend one the home doesn't need, and the sale may not survive it.

220.87 was hard to apply.

The provision has been in the code for years — but using it meant chasing utility data and building the calculation by hand.

THE TRUELOAD APPROACH

A standard load calculation estimates demand from equipment ratings and demand tables — population averages, calibrated decades ago.

NEC 220.87 takes the other path: establish the home's true peak from the utility's own interval meter data, apply the code's 125% safety factor, and compare the result to what the service can actually deliver.

"The code is estimating when it no longer has to."

HOW IT WORKS

1

Request the data

TrueLoad gives your customer a pre-written email. The utility sends the interval data file to you.

2

Set up the job

Enter the existing service — breaker size and voltage — and the equipment you plan to add.

3

Upload the file

CSV, Excel, or XML. TrueLoad finds the true peak and shows PASS or FAIL instantly.

4

Print the report

A formatted 220.87 calculation, ready to submit with the permit application.

From first project to finished report in **under an hour** — no spreadsheets, no manual math.

In the code book — not a gray area.

NEC 220.87 is an established provision of the National Electrical Code, and measured-demand calculations are gaining acceptance with electrical inspectors. What earns approval is documentation: where the data came from, what peak was observed, and how the calculation was made. A TrueLoad report puts all three in the inspector's hands.

Most homes have room to spare

Average U.S. home peak demand vs. a standard 100 A service

100 A service capacity

24 kW

9.7 kW

average home peak demand

≈ 60 A of headroom

already there, sitting unused

0 6 12 18 24 kW

Peak demand: Lawrence Berkeley National Laboratory analysis of 11,940 U.S. homes (2026). $100\text{ A} \times 240\text{ V} = 24\text{ kW}$.

12 mo

of utility interval data behind every calculation

35,000+

meter readings analyzed per project — automatically

125%

NEC safety factor applied to the observed peak

1 PDF

permit-ready report, printed straight from the browser

See TrueLoad in actionQuestions? support@trueloadapp.com**trueloadapp.com**

Works in any browser — nothing to install

TrueLoad is a service adequacy evaluation aid. This tool does not constitute engineering services, guarantee code compliance, or imply AHJ approval. Final responsibility for code compliance remains with the licensed preparer and the Authority Having Jurisdiction.

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